Version 2

Defense Logistics Agency (DLA)
FY26 Small Business Innovation Research (SBIR) BAA
Release 3
Proposal Submission Instructions

INTRODUCTION

The Defense Logistics Agency's (DLA) mission has three lines of effort the DLA Small Business Innovation Program (SBIP) supports. They include supporting the **NUCLEAR ENTERPRISE** by maintaining nuclear systems readiness, qualifying alternate sources of supply, improving the quality of consumable parts, and increasing materiel availability. **FORCE READINESS & LETHALITY** through Improvements to life cycle performance through technological advancement, innovation, and reengineering, mitigate single points-of-failure that threaten the readiness of weapons systems used by our Warfighters. **SUPPLY CHAIN INNOVATION & ASSURANCE** through improved lead times, reduced lifecycle costs, maintaining a secure and resilient supply chain, providing opportunities for the small business industrial base to enhance supply chain operations with technological innovations. Lastly supply chain assurance securing the microelectronics supply chain, development of a domestic supply chain for rare earth elements, the adoptions of industrial base best practices associated with counterfeit risk reduction.

Proposers responding to a topic in this solicitation must follow all general instructions provided in the Department of War (DoW) SBIR Program solicitation. DLA requirements in addition to or deviating from the DoW Program solicitation are provided in the instructions below.

Proposers are encouraged to thoroughly review the DoW solicitation and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The full DoW solicitation is available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate solicitation cycle.
- Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

Specific questions pertaining to the administration of the DLA SBIR Program and these proposal preparation instructions should be directed to:

Defense Logistics Agency
Small Business Innovation Program (SBIP) Office DLA/J68
Email: DLASBIR2@DLA.mil

PHASE I PROPOSAL GUIDELINES

The DoW SBIR/STTR Innovation Portal (DSIP) is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in the DoW solicitation.

Proposal Coversheet (Volume 1)

The proposal coversheet is prepared on DSIP and in accordance with the DoW solicitation.

Technical Volume (Volume 2)

The technical volume is not to exceed 20 pages and must follow the formatting requirements provided in the DoW SBIR solicitation Any pages submitted beyond the 20-page limit within the

Technical Volume (Volume 2) will not be evaluated. If including a letter(s) of support, they should be included in Volume 5, and they will not count towards the 20-page Volume limit. Any technical data/information that should be in the Volume 2 but is contained in other Volumes will not be considered.

Content of the Technical Volume
Refer to the instructions provided in the DoW Program BAA.

Cost Volume (Volume 3)
The Phase I Base amount must not exceed $100,000. Costs for the Base must be clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3. Proposers must utilize the excel cost volume provided during proposal submission on DSIP.

Please review the updated Percentage of Work (POW) calculation details included in the DoW solicitation. SBIR ONLY: DLA will not accept any deviation to the POW requirements / DLA will occasionally accept deviations from the POW requirements with written approval from the Funding Agreement officer. STTR ONLY: Deviations from the POW requirements are not permitted.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW solicitation for full details on this requirement. Information contained in the CCR will be considered by DLA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoW solicitation for more information.

Additional DLA-specific supporting documents:

- Optional, A qualified letter of support is from a relevant commercial or Government Agency procuring organization(s) working with DLA, articulating their pull for the technology (i.e., what DLA need(s) the technology supports and why it is important to fund it), and possible commitment to provide additional funding and/or insert the technology in their acquisition/sustainment program.
- Letters of support shall not be contingent upon award of a subcontract.

The standard formal deliverables for a Phase I are the:

- Plan of Action and Milestones (POAM) with sufficient detail for monthly project tracking.
- Initial Project Summary: one-page, unclassified, non-sensitive, and non-proprietary summation of the project problem statement and intended benefits (must be suitable for public viewing).
- Monthly Status Report. A format will be provided at the Post Award Conference (PAC).
- The Technical Point of Contact (TPOC) and the Program Manager (PM) will determine a meeting schedule at the PAC. Phase I awardees can expect monthly (or more frequent) project reviews.
- Draft Final Report including major accomplishments, business case analysis, commercialization strategy, transition plan with timeline and proposed path forward for Phase II.
- Final Report including major accomplishments, business case analysis, commercialization strategy and transition plan with timeline, and proposed path forward for Phase II.
- Final Project Summary (one-page, unclassified, non-sensitive and non-proprietary summation of project results, high resolution photos or graphics intended for public viewing).
- Applicable patent documentation.

- Other deliverables as defined in the Phase I Proposal.
- Phase II Proposal is optional at the Phase I Awardee's discretion (as applicable).

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoW Program BAA.

PHASE II PROPOSAL GUIDELINES
Per SBA Policy Directive SBIR Phase II Proposal guidance, all Phase I awardees are permitted to submit a Phase II proposal for evaluation and potential award selection, without formal invitation. Details on the due date, format, content, and submission requirements of the Phase II proposal will be provided by the DLA SBIP Program Management Office (PMO) on/around the midway point of the Phase I period of performance. Only firms who receive a Phase I award may submit a Phase II proposal.

DLA will evaluate and select Phase II proposals using the same criteria as Phase I evaluation. Funding decisions are based upon the results of work performed under a Phase I award, the Scientific & Technical Merit, Feasibility, and Commercial Potential of the Phase II proposal; Phase I final reports may be reviewed as part of the Phase II evaluation process. The Phase II proposal should include a concise summary of the Phase I effort including the specific technical problem or opportunity addressed and its importance, the objective of the Phase I effort, the type of research conducted, findings or results of this research, and technical feasibility of the proposed technology.

Due to limited funding, DLA reserves the right to limit awards under any topic and only proposals considered to be of superior quality will be funded.

Phase II Proposals should anticipate a combination of any or all the following deliverables:

- Plan of Action and Milestones (POAM) with sufficient detail for monthly project tracking.
- Initial Project Summary: one-page, unclassified, non-sensitive, and non-proprietary summation of the project problem statement and intended benefits (must be suitable for public viewing).
- Monthly Status Report. A format will be provided at the PAC.
- Meeting schedule to be determined by the Technical Point of Contact (TPOC) and PM at the PAC.
- Phase II awardees expect Monthly (minimum) Project Reviews (format provided at the PAC).
- Draft Final Report including major accomplishments, commercialization strategy and transition plan and timeline.
- Final Report including major accomplishments, commercialization strategy, transition plan, and timeline.
- Final Project Summary (one-page, unclassified, non-sensitive and non-proprietary summation of project results, non-proprietary high-resolution photos, or graphics intended for public viewing).
- Applicable patent documentation.

- Other deliverables as defined in the Phase II Proposal.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
The Small Business Innovation and Economic Security Act Section 7 mandates agencies to offer TABA. DLA will provide up to $6,500 per Phase I project and up to $50,000 per Phase II project OR up to $25,000 for the initial Phase II award and up to an additional $25,000 for one sequential Phase II award, for a total of $50,000 per project.

As identified in 15 U.S.C.§638(q)(1), SBCs may use TABA funding to assist SBIR/STTR projects with:
(1) Making better technical decisions;
(2) Solving technical problems;
(3) Minimizing associated technical risks;
(4) Developing and commercializing new SBIR/STTR commercial products and processes;
(5) Screening for potential foreign involvement in technology development or commercialization activities.

DLA may require SBCs that receive TABA funding to submit a report detailing the results and benefits of the service received and may perform targeted TABA funding reviews at the discretion of the Program Manager.

In addition to the requirements outlined in the DoW solicitation, TABA requests submitted to DLA must fall within the eligible uses as identified in 15 U.S.C.§638(q)(2) and described assistance areas:
- Access to a network of scientists and engineers engaged in a wide range of technologies
- Access to technical and business literature through online databases
- Product sales
- Intellectual property protections
- Cybersecurity assistance, including Cybersecurity Maturity Model Certification (CMMC)
- Market research
- Market validation
- Regulatory plans development
- Manufacturing plans development
- Hiring new staff
- Augment staff
- Direct staff to conduct or participate in training activities

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the DoW solicitation.
Use of Support Contractors in the Evaluation Process

Only government personnel with active non-disclosure agreements will officially evaluate proposals.

Non-government technical consultants (subsequently referred to as "consultants") to the government may review and provide support in proposal evaluations during source selection.

Consultants may have access to the offeror's proposals, may be utilized to review proposals, and may provide comments and recommendations to the government's decision makers. Consultants will not establish final assessments of risk and will not rate or rank offerors' proposals. They are also expressly prohibited from competing for DLA SBIR awards in the SBIR topics they review and/or on which they provide comments to the government.

All consultants are required to comply with procurement integrity laws. Consultants will not have access to proposals or pages of proposals that are properly labeled by the offerors as "FEDONLY." Pursuant to FAR 9.505-4, DLA contracts with these organizations include a clause which requires them to:

(1) Protect the offerors' information from unauthorized use or disclosure for as long as it remains proprietary and

(2) Refrain from using the information for any purpose other than that for which it was furnished. In addition, DLA requires the employees of those support contractors that provide technical analysis to the SBIR/STTR Program to execute non-disclosure agreements. These agreements will remain on file with the DLA SBIP PMO.

Consultants will be authorized access to only those portions of the proposal data and discussions that are necessary to enable them to perform their respective duties. In accomplishing their duties related to the source selection process, employees of the organizations may require access to proprietary information contained in the offerors' proposals.

All proposals will be evaluated in accordance with the evaluation criteria listed in the DoW SBIR Program BAA. DLA will evaluate and select Phase I and Phase II proposals using scientific review criteria based upon technical merit and other criteria as discussed in this Announcement document.

- DLA reserves the right to award none, one, or more than one contract under any topic.
- DLA is not responsible for any money expended by the offeror before award of any contract.
- Due to limited funding, DLA reserves the right to limit awards under any topic.
- Only proposals that DLA considers to be "Highly Acceptable" will be funded.

Please note that potential benefit to DLA will be considered throughout all the evaluation criteria and in the best value trade-off analysis. When combined, the stated evaluation criteria are significantly more important than cost or price.

It cannot be assumed that reviewers are acquainted with the firm or key individuals or any referenced experiments. Technical reviewers will base their conclusions only on information contained in the proposal. Relevant supporting data such as journal articles, literature, including government publications, etc., should be listed in the proposal and will count toward the applicable page limit.

Final Selection may require an oral presentation. This may include an in-person or virtual meeting. The two-part evaluation process is explained below:

Part I: The evaluation of the Technical Volume will utilize the Evaluation Criteria provided in the DoW SBIR BAA. Once the initial evaluations are complete, all offerors will be notified as to whether they were selected to present the slide deck portion of their proposal within 60 days of the BAA close date. Only proposals receiving a "Highly Acceptable" rating will receive an invitation to present orally.

Part II: If selected for an oral presentation, offerors shall submit a slide deck not to exceed 15 PowerPoint slides to DLASBIR2@dla.mil.

- There are no set format requirements other than the 15-page maximum page length.
- It is recommended (but not required) that more detailed information is included in the technical volume and higher-level information is included in the slide deck.

Selected offerors will receive an invitation to present a slide deck (15-minute presentation time / 15-minute question and answer) in a technical question and answer forum to the DLA evaluation team via electronic media. This presentation will be evaluated by a panel against the criteria listed above and your overall presentation. DLA will evaluate the presentation for business acumen, and core business capabilities (customer engagement / presentation skills). The rating of the presentation will be a Go/No-Go rating.

Notification of the Go/No-Go rating decision will occur within 5 days of the presentation. Input on technical aspects of the proposals may be solicited by DLA from consultants and advisors who are bound by appropriate non-disclosure requirements.

The SBIP PMO will distribute selection and non-selection email notices to all firms who submit a SBIR/STTR proposal to DLA. The email will be distributed to the "Corporate Official" and "Principal Investigator" listed on the proposal coversheet. DLA cannot be responsible for notification to a company that provides incorrect information or changes such information after proposal submission.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 days of the closing date of the topic. DLA will provide written feedback to unsuccessful offerors regarding their proposals on the non-selection notification. Only firms that receive a non-selection notification are eligible for written feedback.
Refer to the DoW solicitation for procedures to protest the Announcement.

As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: The contracting officer identified on the subject award.

AWARD AND CONTRACT INFORMATION

Typically, the contract period of performance for Phase I should be up to 12 months and the award should not exceed $100,000. However, each topic may have a different threshold. The DLA Contracting Office utilizes a Firm-Fixed-Price (FFP) Contract for DLA Phase I Projects.

The expected budget for Phase II should not exceed $1,000,000 unless approved by the DLA Program Manager, and the duration should not exceed 24 months. Proposals for more than $1,000,000 will not be considered without written PM approval. The DLA Contracting Office utilizes a FFP/Level-of-Effort Contract for DLA Phase II Projects.

Proposals not conforming to the terms of this announcement will not be considered. DLA reserves the right to limit awards under any topic, and only those proposals that DLA determines to be of superior scientific and technical quality will be funded.

DLA reserves the right to withdraw from negotiations at any time prior to contract award.

Post Award, DLA may terminate any award at any time for any reason to include matters of national security (foreign persons, foreign influence or ownership, inability to clear the firm or personnel for security clearances, or other related issues).

Please read the entire DoW Announcement and DLA instructions carefully prior to submitting your proposal. Please go to https://www.sbir.gov/about to read the SBIR/STTR Policy Directive issued by the Small Business Administration.

USE OF FOREIGN NATIONALS (also known as Foreign Persons), GREEN CARD HOLDERS AND DUAL CITIZENS

If proposing to use foreign nationals (also known as foreign persons), they must be green card holders, and/or dual citizens. No Student or Temporary Visa holders will be approved. The offeror must identify the personnel they expect to be involved on this project, the type of visa or work permit under which they are performing, country of origin and level of involvement.

You will be asked to provide additional information during negotiations to verify the foreign citizen's eligibility to participate on a SBIR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)).

Proposals submitted to export control-restricted topics and/or those with foreign nationals, dual citizens, or green card holders listed will be subject to security review during the contract negotiation process (if selected for award).

DLA reserves the right to vet all uncleared individuals involved in the project, regardless of citizenship, who will have access to Controlled Unclassified Information (CUI) such as export controlled information. If the security review disqualifies a person from participating in the proposed work, the contractor may propose a suitable replacement.

In the event a proposed person and/or firm is found ineligible by the government to perform proposed work, the contracting officer will advise the offeror of any disqualifications but is not required to disclose the underlying rationale.

V. EXPORT CONTROL RESTRICTIONS
The technology within most DLA topics is restricted under export control regulations including the International Traffic in Arms Regulations (ITAR) and the Export Administration Regulations (EAR). ITAR controls the export and import of listed defense-related material, technical data and services that provide the United States with a critical military advantage. EAR controls military, dual-use and commercial items not listed on the United States Munitions List or any other export control lists. EAR regulates export-controlled items based on user, country, and purpose. The offeror must ensure that their firm complies with all applicable export control regulations. Please refer to the following URLs for additional information: https://www.pmddtc.state.gov/ and https://www.bis.doc.gov/index.php/regulations/export-administration-regulations-ear.

Most DLA SBIR topics are subject to ITAR and/or EAR. If the topic write-up indicates that the topic is subject to ITAR and/or EAR, your company may be required to submit a Technology Control Plan (TCP) during the contracting negotiation process.

CLAUSE H-08 PUBLIC RELEASE OF INFORMATION (Publication Approval)
Clause H-08 pertaining to the public release of information is incorporated into all DLA SBIR contracts and subcontracts without exception. Any information relative to the work performed by the contractor under DLA SBIR contracts must be submitted to DLA for review and approval prior to its release to the public. This mandatory clause also includes the subcontractor who shall provide their submission through the prime contractor for DLA's review for approval.

FLOW-DOWN OF CLAUSES TO SUBCONTRACTORS
The clauses to which the prime contractor and subcontractors are required to comply include but are not limited to the following clauses:

1) DLA clause H-08 (Public Release of Information),
2) DFARS 252.204-7000 (Disclosure of Information),
3) DFARS clause 252.204-7012 (Safeguarding Covered Defense Information and Cyber Incident Reporting), and
4) DFARS clause 252.204-7020 (NIST SP 800-171 DoW Assessment Requirements). Your proposal submission confirms that any proposed subcontract is in accordance with the clauses cited above and any other clauses identified by DLA in any resulting contract.
5) DFARS Clause 252.223-7999 Ensuring Adequate COVID-19 Safety Protocols for Federal Contractors.

OWNERSHIP ELIGIBILITY
Prior to award, DLA may request business/corporate documentation to assess ownership eligibility as related to the requirements of SBIR Program Eligibility. These documents include, but may not be limited to, the Business License; Articles of Incorporation or Organization; By-Laws/Operating Agreement; Stock Certificates (Voting Stock); Board Meeting Minutes for the previous year; and a list of all board members and officers.

If requested by DLA, the contractor shall provide all necessary documentation for evaluation prior to SBIR award. Failure to submit the requested documentation in a timely manner as indicated by DLA may result in the offeror's ineligibility for further consideration for award.

ADDITIONAL INFORMATION
Classified Proposals
Classified proposals ARE NOT accepted under the DLA SBIR Program. The inclusion of classified data in an unclassified proposal is grounds for the agency to determine the proposal as non-responsive and the proposal not to be evaluated.

Contractors currently working under a classified contract must use the security classification guidance provided under that contract to verify new SBIR proposals are unclassified prior to submission.

Phase I contracts are not typically awarded for classified work. However, in some instances, work being performed on DLA SBIR/STTR contracts will require security clearances. If a DLA SBIR/STTR contract develops into or identifies classified work, the offeror must have a facility clearance, appropriate personnel clearances to perform the classified work and coordinate the DD254 with the Contract Officer and the service owning the classified data.

For more information on facility and personnel clearance procedures and requirements, please visit the Defense Counterintelligence and Security Agency website at: https://www.dcsa.mil/.

Use of Acronyms
Acronyms should be spelled out the first time they are used within the technical volume (Volume 2), the technical abstract, and the anticipated benefits/potential commercial applications of the research or development sections. This will help avoid confusion when proposals are evaluated by technical reviewers.

Communication
All communication from the DLA SBIR/STTR PMO will originate from the DLASBIR2@DLA.mil email address. Please white list this address in your company's spam filters to ensure timely receipt of communications from our office.

All attachments sent via email require encryption. The firm will have to purchase External Certificate Authority (ECA) certificates to send and receive encrypted email if they do not have a Common Access Card (CAC) or Personal Identity Verification (PIV) issued. The cost is approximately $100 per year per user. This will be a Cybersecurity Maturity Model Certification (CMMC) requirement for all future contracts.

ORGANIZATIONAL CONFLICTS OF INTEREST (OCI)
The basic OCI rules for contractors which support development and oversight of SBIR topics are covered in FAR 9.5 as follows (the offeror is responsible for compliance):

(1) The contractor's objectivity and judgment are not biased because of its present or planned interests which relate to work under this contract.

(2) The contractor does not obtain unfair competitive advantage by virtue of its access to non-public information regarding the government's program plans and actual or anticipated resources.

(3) The contractor does not obtain unfair competitive advantage by virtue of its access to proprietary information belonging to others.

All applicable rules under the FAR Section 9.5 apply.

If you, or another employee in your company, developed or assisted in the development of any SBIR requirement or topic, please be advised that your company may have an OCI. Your company could be precluded from an award under this BAA if your proposal contains anything directly relating to the development of the requirement or topic. Before submitting your proposal, please examine any potential OCI issues that may exist with your company to include subcontractors and understand that if any exist, your company may be required to submit an acceptable OCI mitigation plan prior to award.
PHASE III GUIDELINES & INSTRUCTIONS

Phase III is any proposal that derives from, extends, or completes a transition from a Phase I or II project. Phase III proposals will be accepted after the completion of Phase I and/or Phase II projects.

There is no specific funding associated with Phase III, except Phase III is not allowed to use SBIR/STTR coded funding. Any other type of funding is allowed.

Phase III proposal submission. Phase III proposals are emailed directly to DLASBIR2@dla.mil. The PMO team will set up evaluations and coordinate the funding and contracting actions depending on the outcome of the evaluations. A Phase III proposal should follow the same format as Phase II for the content, and format. There are, however, no limitations to the amount of funding requested, or the period of performance. All other guidelines apply. More specific instructions may be available when a firm submits a Phase III proposal.

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DLA SBIR DoW 2026 BAA
Release 3
Phase I Topic Index

DLA26BZ03-NV011 TITLE: Digital Twin of the Organization for Enhanced Mission Readiness

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Sustainment & Logistics; Mission Readiness & Disaster Preparedness; Advanced Materials

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE:
The Defense Logistics Agency (DLA) seeks to develop and implement a dynamic workforce digital twin to enhance organizational effectiveness, support mission-critical decisions, and accelerate the adoption of advanced technologies. As Artificial Intelligence (AI) fundamentally changes the workforce of the future, the Department must achieve a 10x productivity increase to maintain strategic overmatch and compete effectively with near-peer adversaries and the workforce plays a critical role in achieving this scale. The primary objective is to create a Digital Twin (DT) of the organization capable of generating synthetic data and enabling leaders to react quickly to dynamic changes. This tool will inform key workforce-related decisions, improve how work gets done, and allow for robust scenario-planning for core mission activities under unpredictable global conditions.

DESCRIPTION:
A digital twin of the organization is a critical enabler for the DLA, empowering leaders to model and test workforce scenarios, optimize processes, and achieve the exponential efficiency gains required for modern great power competition. Current static models are insufficient for the agility needed today. The proposed solution will move beyond legacy systems by integrating real-time data to provide dynamic insights into workforce capabilities, productivity limitations, and the potential impacts of organizational restructuring.

Crucially, this organizational DT must possess the capability to generate synthetic data to simulate complex, unforeseen scenarios, allowing the enterprise to react rapidly to supply chain disruptions, geopolitical shifts, or internal surges. By modeling the workforce of the future—one that is heavily augmented by AI—this effort will explore innovative data sources and functionalities to address critical questions:

- Strategic Competition & Productivity: How can we leverage the DT to identify structural and process pathways to a 10x productivity increase, countering the scale of adversaries like China?

- Surge Deployment & Rapid Reaction: How ready is the workforce for a sudden surge scenario? Utilizing synthetic data generation, how can we proactively test personnel moves to backfill deployed staff without mission degradation?
- AI Integration & Organizational Structure: What are the root causes of manual work, and what activities must be stopped or automated by AI? How does the organizational structure need to shift to maximize human-machine teaming and efficiently integrate new mission sets?

Research and Development (R&D) efforts for this topic should demonstrate a technical feasibility that has not been fully established and must be judged to be at a Technology and/or Manufacturing Readiness Level (TRL/MRL) between 3 and 6 to be considered for funding.

PHASE I:
Not to exceed a duration of 12 months and a cost of $100,000. Phase I will focus on exploring the value of a digital workforce twin by delivering a proof-of-concept. This phase includes:

1. Identifying and ranking high-impact use cases (e.g., surge deployment in human capital, order-to-cash in finance, human-machine teaming integration).
2. Identifying necessary data sources, articulating a data acquisition and synthetic data generation plan, and defining success criteria.
3. Developing a business case on the expected ROI of the tool and a product roadmap for a Minimum Viable Product (MVP).

An alignment or collaboration with a relevant DLA Component organization (e.g., DLA J1, J6, J3, and J7) is highly desirable.

PHASE II:
Phase II will consist of establishing a pilot-scale process to build, test, and validate a functioning Minimum Viable Product (MVP) of the digital workforce twin solution. This will be conducted in a series of milestone-driven agile sprints to rapidly deliver incremental value and incorporate user feedback. The goal is to produce a functioning tool that can simulate scenarios via synthetic data, scale to other use cases, and refine the business case for further investment based on demonstrated mission value in increasing productivity. Collaboration with a relevant DoW Component organization is highly desirable.

PHASE III DUAL USE APPLICATIONS:
At this point, no specific funding is associated with Phase III. The progress and relationships developed in Phases I and II should position the solution for transition to production in support of DoW orders. The underlying technology can be adapted for a wide range of commercial and public sector applications, including corporate strategic planning, human resource management in large enterprises, and organizational design for complex, non-DoW government agencies navigating the AI transition. The goal is the organic growth of the business by applying the solution to both defense and commercial markets.

REFERENCES:

1. *Digital Twins – when and why to use one*
2. *What is Digital Twin technology?*
3. *Digital twins: How to build the first twin*

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KEYWORDS:
Digital Twin, Workforce, Organization, Artificial Intelligence, Synthetic Data, Scenario Planning, Great Power Competition, Productivity, Logistics, Mission Readiness

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DLA26BZ03-NV012 TITLE: AI-Powered Tool for Automated Evaluation of Vendor Economic Dependency

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Information Systems

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE**:** Develop an innovative, AI-driven tool to automate the assessment of economic dependency for vendors within the Defense Logistics Agency's (DLA) supply chain. This capability will enable DLA to proactively identify relationships and analyze potential related-party transactions and economic dependencies in compliance with federal accounting standards and audit recommendations, including specifically Statements of Federal Financial Accounting Standards 47, thereby enhancing supply chain resilience, financial stewardship, and audit compliance.

DESCRIPTION: DLA's global mission relies on a vast and diverse industrial base. Ensuring financial transparency and mitigating supply chain risk requires a comprehensive understanding of the economic relationships between DLA and its key suppliers. Current methods for this analysis are manual, time-consuming, and cannot effectively scale across thousands of vendors and millions of transactions.

This SBIR topic seeks the development of an AI-powered tool to automate this process. The desired solution would integrate with DLA's business systems to identify significant vendor relationships and automatically retrieve publicly available financial data (e.g., from SEC filings). Using this data, the tool will apply a defined criterion for economic dependency (e.g., percentage of a vendor's revenue derived from DLA) to flag potential related parties. The solution should also be capable of assessing risk based on contract type (e.g., cost-reimbursement vs. fixed-price). The final tool must be designed to operate in a secure government environment and provide auditable, traceable results.

PHASE I: Conduct a feasibility study to demonstrate the core concepts. This includes developing a proof-of-concept tool that can successfully identify a universe of vendors from sample contract data, retrieve public financial information from sources like the SEC's EDGAR system, commercial public 10K reports, SAM.gov, and apply the economic dependency criteria. The study must address SFFAS 47. The final report should include the prototype design, preliminary results, established golden dataset to base subsequent review and analysis upon, and a detailed plan for a Phase II effort. The Phase I award will not exceed $100,000 over a 12-month period.

PHASE II: Develop a scalable prototype of the AI tool within a government-approved development environment that fits DLA tech stack. The prototype must demonstrate the ability to process a large volume of vendor and contract data, accurately retrieve external information to support the evaluation of relationships and economic dependencies, assess and map/categorize risk and materiality, and provide transparent evidence for DLA assertions and/or disclosures with deep reasoning anchored to SFFAS 47.

Phase II will include rigorous testing to validate the accuracy and efficiency of the tool and will produce a detailed transition plan for integration into DLA's operational environment. The Phase II award will not exceed $1,000,000 over a 24-month period.

PHASE III: A successful solution will be transitioned from development to production for operational use within DLA's environment and technology stack to support ongoing audit readiness and supply chain risk management. This technology has applicability for other DoW components and federal agencies seeking to automate financial oversight and identify concentration risk within their own supply chains or those using Defense Capital Working Funds (DCWF) to support investment analysis, strategic supply chain management, budgetary projections, and financial compliance, where identifying and understanding economic dependencies on the DLA and DoW is critical to fiscal stewardship.

REFERENCES:

1. OMB Memorandum M-24-10: Advancing Governance, Innovation, and Risk Management for Agency Use of Artificial Intelligence (March 2024)
2. NDAA FY2024, Section 1542: Strategy and Guidance for the Use of AI in Defense Business Systems (December 2023)
3. NDAA FY2024, Section 1002: Plan for Attainment of Unqualified Audit Opinion (December 2023)
4. Executive Order 14110: Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (October 2023)
5. SFFAS 62: Transitional Amendment to SFFAS 54, Leases (May 2023)
6. NDAA FY2021, Section 881: Promotion of Artificial Intelligence Capabilities (January 2021)
7. NDAA FY2020, Section 847: Mitigating Risks Related to Foreign Ownership, Control, or Influence (FOCI) (December 2019)
8. SFFAS 54: Leases (April 2018)
9. OMB Circular A-123: Management's Responsibility for Enterprise Risk Management and Internal Control (July 2016)
10. SFFAS 47: Reporting Entity (December 2014, Effective FY2018)

KEYWORDS: Artificial Intelligence, SFFAS, Audit Readiness, Supply Chain Risk Management, Financial Analysis, Related Party, Economic Dependency, Natural Language Processing

Version 2

DLA26BZ03-NV013 TITLE: Next-Generation Non-Lithium Battery Technology for Safe, Extreme-Environment, High-Performance Resilient Military Logistics and Field Operations

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technology (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Battery Materials; Electrochemical Energy Storage; Advanced Manufacturing; Hazardous Materials Reduction; Next Generation Battery Technologies

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE:
Develop and demonstrate a **non-lithium battery cell and/or prototype battery pack** capable of supporting BB-2590/U-class applications, with emphasis on safety, transportability, and domestic manufacturability, while achieving **mission-relevant performance thresholds**.

DESCRIPTION:
DLA's current battery portfolio, including the widely fielded BB-2590/U rechargeable battery (NSN: 6140-01-490-4316), relies heavily on lithium-ion chemistries with significant exposure to foreign (including Chinese-origin) supply chains. Section 154 of the FY2024 NDAA restricts DoD procurement from certain foreign sources beginning in October 2027, creating an urgent need for compliant alternatives.

Beyond supply chain risk, lithium-based batteries present operational challenges including hazardous material classification for transport, thermal runaway risks, limited long-duration storage performance, and handling complexity in forward-deployed environments.

DLA requires a non-lithium battery solution that can support BB-2590/U-class applications while improving:

- Safety: Reduced or eliminated thermal runaway risk
- Transportability: Less restrictive hazardous material classification and simplified logistics handling
- Storage Readiness: Low self-discharge and long-duration storage (≥12–24 months)
- Operational Performance: Reliable function across -30°C to +60°C environments
- Durability: Resistance to shock, vibration, and field conditions
- Supply Chain: Fully domestic or NDAA-compliant sourcing

Solutions may not need to exceed lithium-ion performance in all areas but must demonstrate clear operational advantages in safety, logistics, or sustainment.

Performance Targets:
Proposed solutions should aim to meet or approach:

- Energy Density: ≥150 Wh/kg (threshold), ≥200 Wh/kg (objective)
- Cycle Life: ≥500 cycles to 80% capacity
- Storage Life: ≥12 months with minimal degradation
- Operating Temperature: -30°C to +60°C
- Safety: No thermal runaway under standard abuse conditions
- Transport: Reduced hazardous classification compared to Li-ion (if applicable)

- MIL-STD environmental qualification testing covering temperature, vibration, and mechanical shock
- Preliminary safety qualification aligned with MIL-PRF-32383/3
- Validated domestic bill of materials demonstrating NDAA-compliant manufacturability
- Commercialization plan and domestic manufacturing roadmap with cost-per-unit projections at production volume

Scope
Phase I (6 months, up to $100K)
Focus on **cell-level feasibility and trade-space validation**:

- Demonstrate a **non-lithium electrochemical system** with initial performance data
- Characterize:
    - Energy density (Wh/kg, Wh/L)
    - Cycle life projections
    - Voltage profile and discharge characteristics
    - Temperature performance
- Conduct **preliminary safety testing**, including abuse tolerance
- Perform **storage/self-discharge testing**
- Provide:
    - Concept for integration into BB-2590/U-class form factor
    - Initial domestic supply chain assessment
    - Phase II development plan

Phase II (18 months, up to $1M)
Focus on **prototype demonstration and transition readiness**:

- Develop a **prototype battery module or pack** aligned with BB-2590/U electrical requirements
- Conduct:
    - Environmental testing (temperature, shock, vibration)
    - Extended cycle and storage validation
    - Safety characterization aligned with military battery standards
- Deliver:
    - Prototype system demonstration
    - Validated domestic Bill of Materials
    - Manufacturing feasibility and cost model
    - Transition plan including pathway to qualification and DLA procurement

PHASE I:
Phase I Deliverables (up to 6 months, up to $100,000)

The Phase I effort will establish the core technical feasibility of the proposed non-lithium battery chemistry through systematic cell characterization and safety testing. The following deliverables are expected:

Electrochemical Characterization: Prototype cells at a relevant form factor shall be characterized and tested, with performance data reported for energy density in Wh/kg and Wh/L, cycle life to 80% capacity retention, nominal voltage, and operational temperature range across both cold and high-heat conditions.

Safety Testing: Thermal stability characterization shall be conducted to demonstrate the absence of thermal runaway response under representative charge, discharge, and abuse conditions, establishing the fundamental safety advantage of the proposed chemistry over conventional lithium-based systems.

Self-Discharge Assessment: A storage stability assessment shall be conducted to measure self-discharge rates over a defined period, providing quantitative data to evaluate the technology's potential for reducing infrastructure overhead associated with long-term DLA war-reserve and pre-positioned stock storage.

PHASE II:
Phase II Deliverables (18 months, up to $1,000,000)

Building on the Phase I feasibility demonstration, Phase II will advance the technology to a battery pack level and generate the qualification and commercialization data required for DLA procurement consideration. The following deliverables are expected:

- Scaled Battery Pack Prototype: A fully integrated battery pack prototype shall be developed meeting the form factor and electrical specifications of target DLA battery applications, demonstrating the scalability of the Phase I cell technology to a system-level configuration.
- MIL-STD Environmental Testing: The prototype shall be subjected to MIL-STD environmental qualification testing covering operational temperature performance and mechanical vibration, demonstrating suitability for deployment across diverse military platforms and environments.
- Preliminary Safety Qualification: Safety qualification data shall be generated in alignment with applicable MIL-PRF specifications, providing DLA and its program offices with a preliminary compliance baseline for procurement and fielding consideration.
- Validated Bill of Materials: A validated bill of materials shall be developed demonstrating the domestic manufacturability of the battery system, supporting DLA's interest in building a resilient and accessible domestic supply base for next-generation battery technologies.
- Commercialization Plan and Manufacturing Roadmap: A detailed commercialization plan and domestic manufacturing roadmap shall be delivered, including cost-per-unit projections at production volume, outlining the pathway for transitioning the technology from prototype to DLA-procurable supply item and broader commercial markets — including heavy-duty vehicle electrification, maritime autonomous systems, and stationary grid-scale energy storage.

PHASE III DUAL USE APPLICATIONS:
This technology has strong dual-use potential across both defense and commercial markets.

Defense Applications:

- Man-portable communications systems
- Unmanned systems (UGV/UAS)
- Expeditionary sensors and electronic warfare systems
- Pre-positioned war reserve energy storage

Commercial Applications:

- Industrial and warehouse robotics
- Remote sensing and IoT systems
- Medical and safety-critical portable devices
- Backup and stationary energy storage systems
- Maritime and off-grid energy solutions

Version 2

The emphasis on **safety, long storage life, and simplified logistics** positions non-lithium technologies as attractive alternatives in markets where lithium-ion risks or transport constraints are limiting factors.

REFERENCES:

1. National Defense Authorization Act for Fiscal Year 2024, Pub. L. No. 118-31, § 154, 137 Stat. 136, 163–164 (2023). congress.gov https://www.congress.gov/bill/118th-congress/house-bill/2670/text
2. Federal Consortium for Advanced Batteries. (2021, June). National blueprint for lithium batteries 2021–2030. U.S. Department of Energy. energy.gov https://www.energy.gov/sites/default/files/2021-06/FCAB%20National%20Blueprint%20Lithium%20Batteries%200621_0.pdf
3. U.S. Department of Defense. (2011). MIL-PRF-32383/3: Performance Specification Sheet: Battery, Rechargeable, Sealed, BB-2590()/U, BB-390()/U, and BB-3590()/U. ASSIST-QuickSearch. https://assistca.dla.mil/online/doc_analysis/doc_info_general.cfm?ident_number=277791

KEYWORDS:
Batteries, Battery Technology, Non-Lithium Battery, Military Logistics, Field Operations